



National University of Modern Languages, Islamabad

LAB 6

Submitted To:	Sir Shahid
Submitted By:	Tayyaba Ejaz
Roll Number:	2751
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1. Lab Task for Attendance App

Create a Flutter project and integrate the Hive and hive_flutter packages. Initialize Hive properly in the main.dart file and ensure that the application is ready to use local storage before the user interface is displayed. You must define a custom Student model class containing at least two attributes: name and age. Since Hive does not directly understand custom Dart objects, you are required to create a TypeAdapter for the Student class using Hive annotations such as @HiveType and @HiveField, and generate the adapter using the appropriate build runner command. The adapter must be registered before storing any student data.

The application should provide a simple user interface containing input fields that allow the user to enter a student's name and age, along with a button labeled "Save Student". When the user presses the button, a Student object should be created and saved into a Hive box named appropriately for storing students. Proper input handling should be applied to ensure that the entered data is stored correctly.

In addition to storing data, the application must also retrieve and display all saved student records from the Hive box. The stored students should be shown in a list format using a ListView, where each item clearly displays the student's name and age. The displayed data must persist across app restarts, demonstrating successful offline storage.

1.1 Code

1.1.1 Model: Student.dart

```
import 'package:hive/hive.dart';
```

```
part 'student.g.dart';
```

```
@HiveType(typeId: 0)
```

```
class Student {
```

```
  @HiveField(0)
```

```
  String name;
```

```
  @HiveField(1)
```

```
int age;
```

```
Student({required this.name, required this.age});
```

```
}
```

1.1.2 Student.g.dart

```
// GENERATED CODE - DO NOT MODIFY BY HAND
```

```
part of 'student.dart';
```

```
//
```

```
*****
```

```
// TypeAdapterGenerator
```

```
//
```

```
*****
```

```
class StudentAdapter extends TypeAdapter<Student> {
```

```
  @override
```

```
  final int typeId = 0;
```

```
  @override
```

```
  Student read(BinaryReader reader) {
```

```
    final numOfFields = reader.readByte();
```

```
    final fields = <int, dynamic>{
```

```
      for (int i = 0; i < numOfFields; i++) reader.readByte(): reader.read(),
```

```
    };
```

```
    return Student(
```

```
      name: fields[0] as String,
```

```
      age: fields[1] as int,
```

```
);  
}
```

```
@override  
void write(BinaryWriter writer, Student obj) {  
  writer  
    ..writeByte(2)  
    ..writeByte(0)  
    ..write(obj.name)  
    ..writeByte(1)  
    ..write(obj.age);  
}
```

```
@override  
int get hashCode => typeId.hashCode;
```

```
@override  
bool operator ==(Object other) =>  
  identical(this, other) ||  
  other is StudentAdapter &&  
    runtimeType == other.runtimeType &&  
    typeId == other.typeId;  
}
```

1.1.3 Main: main.dart

```
import 'package:flutter/material.dart';  
import 'package:hive_flutter/hive_flutter.dart';  
import 'student.dart';
```

```
void main() async {  
  WidgetsFlutterBinding.ensureInitialized();  
  
  // Initialize Hive  
  await Hive.initFlutter();  
  
  // Register adapter  
  Hive.registerAdapter(StudentAdapter());  
  
  // Open box  
  await Hive.openBox<Student>('studentsBox');  
  
  runApp(const MyApp());  
}  
  
class MyApp extends StatelessWidget {  
  const MyApp({super.key});  
  
  @override  
  Widget build(BuildContext context) {  
    return MaterialApp(  
      home: HomePage(),  
    );  
  }  
}
```

```
class HomePage extends StatefulWidget {  
    @override  
    State<HomePage> createState() => _HomePageState();  
}
```

```
class _HomePageState extends State<HomePage> {  
    final nameController = TextEditingController();  
    final ageController = TextEditingController();  
  
    final studentsBox = Hive.box<Student>('studentsBox');  
  
    void saveStudent() {  
        final name = nameController.text.trim();  
        final ageText = ageController.text.trim();  
  
        if (name.isEmpty || ageText.isEmpty) return;  
  
        final age = int.tryParse(ageText);  
        if (age == null) return;  
  
        final student = Student(name: name, age: age);  
  
        studentsBox.add(student);  
  
        nameController.clear();  
        ageController.clear();  
    }  
}
```

```
    setState(() {});  
  }
```

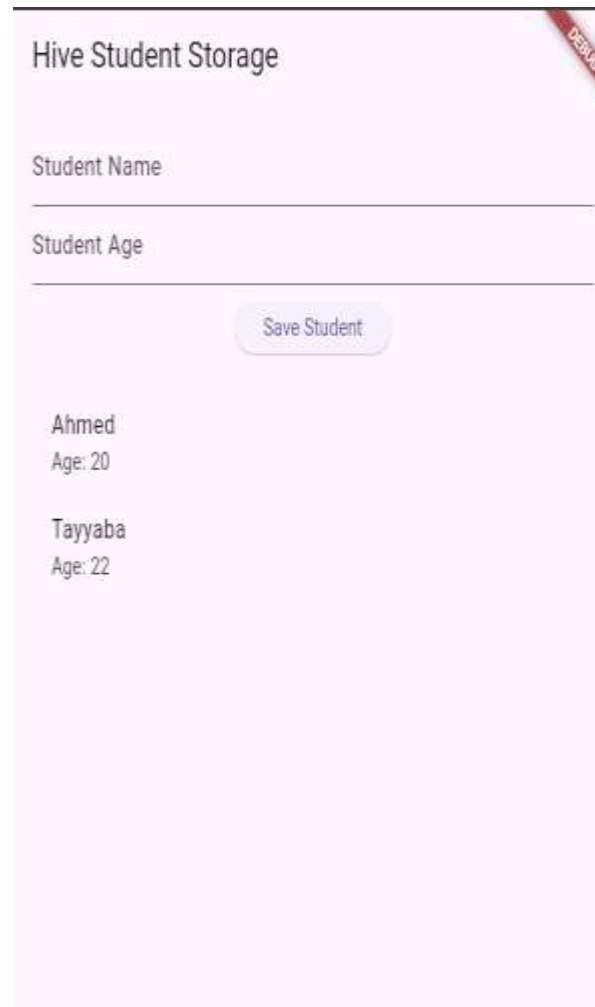
```
@override
```

```
Widget build(BuildContext context) {  
  return Scaffold(  
    appBar: AppBar(title: Text('Hive Student Storage')),  
    body: Padding(  
      padding: const EdgeInsets.all(16),  
      child: Column(  
        children: [  
          TextField(  
            controller: nameController,  
            decoration: InputDecoration(labelText: 'Student Name'),  
          ),  
          TextField(  
            controller: ageController,  
            decoration: InputDecoration(labelText: 'Student Age'),  
            keyboardType: TextInputType.number,  
          ),  
          SizedBox(height: 10),  
          ElevatedButton(  
            onPressed: saveStudent,  
            child: Text("Save Student"),  
          ),  
          SizedBox(height: 20),  
          Expanded(  
            child: Column(  
              children: [
```



```
child: ValueListenableBuilder(  
  valueListenable: studentsBox.listenable(),  
  builder: (context, box, _) {  
    return ListView.builder(  
      itemCount: box.length,  
      itemBuilder: (context, index) {  
        final student = box.get(index);  
        return ListTile(  
          title: Text(student!.name),  
          subtitle: Text("Age: ${student.age}"),  
        );  
      },  
    );  
  },  
),  
],  
),  
);  
}
```

1.2 Screenshots



The screenshot shows a web application titled "Hive Student Storage". It features a light pink background. At the top right, there is a red ribbon icon with the word "Delete" written on it. Below the title, there are two input fields: "Student Name" and "Student Age", each followed by a horizontal line. Below these fields is a rounded rectangular button labeled "Save Student". At the bottom of the form, there are two entries of student data: "Ahmed" with "Age: 20" and "Tayyaba" with "Age: 22".

Marked two students present
